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From rule-based to generative AI: a systematic review of algorithmic evolution and multimodal fusion in VR interview training systems

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The interview performance is crucial for employment success. The combination of artificial intelligence and virtual reality provides job seekers with a safe and repeatable immersive training environment. Following PRISMA 2020 guidelines, we conducted a systematic literature review of immersive virtual reality interview training systems across IEEE Xplore, ACM Digital Library, Web of Science, and Google Scholar from 2015 to 2025. Based on our inclusion and exclusion criteria, 23 unique studies were identified for full synthesis. We find that studies increased after 2020, with over half published between 2023 and 2025. This growth was driven by hardware maturation around 2019, and the rise of generative artificial intelligence, notably large language models such as ChatGPT since late 2022. Notably, our findings reveal a technological evolution centered on the adaptive intervention loop of interaction, perception, and decision. Starting with the interaction module initiating interview stimuli, the system utilizes the perception module to capture real-time multimodal data integrating speech, behaviors (e.g., eye movements and facial expressions), and physiological signals (e.g., heart rate and electrodermal activity), allowing the algorithm module to perform intelligent evaluation and decision-making, which in turn drives the interaction module to execute dynamic interventions. Progressing from rule-based systems and machine learning to generative artificial intelligence and large language models, this evolution has significantly enhanced the systems' interactivity, personalization, and adaptive feedback capabilities. Furthermore, we identify challenges in hardware costs, algorithmic biases, and privacy security, providing a reference for the future development of a more equitable and efficient next-generation intelligent interview system.

KEYWORDS
algorithmic evolution, artificial intelligence algorithms, generative artificial intelligence, job interview training, multimodal data fusion, virtual reality

1 Introduction

Over the past century, employment interviews have remained the primary and most critical tool for recruitment (Posthuma et al., 2002). In an increasingly competitive labor market, job seekers often go through multiple rounds of evaluation before receiving an offer, making interview performance a key factor in career success. However, traditional interview training such as mock interviews, mirror self-practice, and expert coaching has notable limitations. Expert coaching is costly and difficult to scale up, while the mock interview scenarios often fail to effectively replicate the stressful environment of real interviews. Furthermore, the training process often

lacks the personalized adaptation needed to meet the diverse needs of different groups or specific industry requirements.

The value of experiential learning is reflected in the wisdom of the Chinese philosopher Xunzi: "Tell me and I forget, teach me and I may remember, involve me and I learn." This quote reveals that compared to purely theoretical instruction, practice-based immersive training is far more effective for mastering skills. This is especially applicable to social interactions like job interviews, where real-time responsiveness is pivotal. Virtual Reality (VR) technology can create highly realistic interview scenarios with immersive environment, allowing job seekers to practice repeatedly in a safe and controlled environment. Recent empirical studies confirm that such systems effectively alleviate physiological and behavioral anxiety (Geng et al., 2024), while also demonstrably enhancing communication fluency and professional confidence (Chen et al., 2025). Meanwhile, Artificial Intelligence (AI) technology empowers such systems with personalized feedback and real-time evaluation, including automated interview question generation (Amarasena et al., 2024), interview performance assessment (Nofal et al., 2025), and visual feedback dashboards (Adiani et al., 2022). The integration of VR and AI provides an effective approach to interview training, bridging the gap between theoretical preparation and practical performance.

In recent years, research on AI-driven VR interview training systems has been increasing, leading to various multi-dimensional reviews within academia. Based on their focuses, existing reviews can be categorized into the following three types: (1) The frontier AI applications in VR. Following the rise of Generative Artificial Intelligence (GenAI) and Large Language Models (LLMs), Tortora et al. (2025) and Rahimi et al. (2025) have explored how they bring real-time text, image, and dynamic environment generation capabilities to VR, highlighting its profound transformation of content co-creation and immersion enhancement. Although these studies capture the most cutting-edge technical trends, their focus is more concentrated on the content generation. (2) Another focus is the educational training and social interaction within VR environments. Checa and Bustillo (2020) and Oyelere et al. (2020) focused on the design trends and educational effectiveness of immersive VR serious games, analyzing how virtual environments enhance skill acquisition by increasing learner motivation. Shifting from general gameplay to relational dynamics, Kyrlitsias and Michael-Grigoriou (2022) provided an in-depth exploration of the psychological foundations and social presence underlying user interactions with agents and avatars within immersive virtual environments. These reviews provide strong theoretical support for behavior learning. However, they offer little discussion on the algorithmic evolution within the specific domain of job interview training. (3) Some reviews specifically address AI-driven VR job interview training systems. Chang et al. (2024) measured the overall effectiveness of embedding autonomous AI agents inside VR training environments. Firnanda et al. (2025) focused on multimodal AI in automated interview training, providing a deep dive into how integrating text, speech, and nonverbal behavior analysis enhances candidate performance improvement. Sanjana et al. (2025) mapped the transition of intelligent interviewing systems from early rule-based setups to modern AI-enhanced platforms. However, their review specifically targets technical and professional skills, and their analysis remains largely at the level of broad technical categorization without a deep analysis of the algorithms.

Existing reviews rarely isolate job interview training as a dedicated domain, often mixing it with general social interaction training. Furthermore, their analyses typically stay at broad technical categorizations to evaluate system efficacy, lacking a deep analysis of longitudinal algorithmic evolution paths and their underlying mechanics. Discussions on multimodal data fusion also remain highly fragmented. To address these limitations, this review provides a systematic examination of how AI drives the transition of immersive VR interview systems from "static presets" to "dynamic, intelligent interactions" through the longitudinal lens of algorithmic evolution. We fill this gap by focusing strictly on interview training in immersive VR environments, tracing the evolution of AI algorithms from rule-based systems and machine learning models to modern GenAI, and providing a deep analysis of their functional roles. Furthermore, this paper provides an in-depth analysis of multimodal data fusion and outlines current challenges regarding technical efficacy, evaluative validity, and ethical fairness. Therefore, we synthesized a decade of empirical research on AI-driven VR interview training systems. This systematic review aims to answer the following research questions:

1. RQ1: How does AI drive VR interview training systems to evolve from early preset scripts to context-awareness and adaptive interaction in the current literature?
2. RQ2: How does the current literature describe the multimodal data fusion used to achieve a comprehensive assessment of participants' external performance and internal states?
3. RQ3: What are the challenges identified in the current literature for AI-driven VR interview training systems?

2 Methods

2.1 Literature selection

By following the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) 2020 guidelines (Page et al., 2021), we conducted a systematic review of the published literature on Virtual Reality job interview training with Artificial Intelligence technology over the last 10 years. Relevant publications were selected through comprehensive searches in major databases, including IEEE Xplore, ACM Digital Library, Web of Science, and Google Scholar. Specifically, IEEE Xplore and ACM Digital Library were selected as the primary databases to ensure comprehensive coverage of core technical literature in computer science, software engineering, and human-computer interaction. Web of Science was utilized to provide extensive cross-disciplinary coverage, ensuring that critical studies published in behavioral science, psychology, and

organizational training journals were not omitted. Lastly, Google Scholar was integrated as a supplementary source to verify search completeness and capture any remaining uncovered literature.

To query these databases, the search strategy utilized a combination of keywords and Boolean operators to capture studies on AI-driven job interview training systems in VR. Keywords were structured around three conceptual categories: job interview training (e.g., interview training, job interview simulator, job interview evaluator, job interview preparation, job mock interview), VR environments (e.g., virtual reality), and AI technology (e.g., AI, AI-driven). Specifically, the final search query is: ("job interview training" OR "interview training" OR "job interview simulator" OR "job interview evaluator" OR "job interview preparation" OR "job mock interview") AND "virtual reality" AND ("AI" OR "AI-driven"). To ensure transparency and reproducibility, the full search strategy for each database is provided in the Supplementary Material.

The publication range for this review was restricted to studies published between 2015 and 2025. This decade-long timeframe was selected to encompass the complete lifecycle of modern consumer immersive VR, which took off around 2016 with the market releases of the HTC Vive series, Oculus Rift series, and FOVE 0. These early head-mounted displays (HMDs) introduced spatial tracking and integrated eye-tracking capabilities, providing the essential hardware foundation for adaptive interview training systems. Extending the search back to 2015 ensures the inclusion of early prototypes and provides a clear historical baseline for tracking the technological shift from rule-based AI to generative models. The final search across all specified databases was executed on 7 March 2026.

2.2 Eligibility criteria

Eligibility criteria for this systematic review were structured using the PICOS (Population, Intervention, Comparator, Outcome, and Study Design) taxonomy in accordance with PRISMA recommendations, with specific criteria established for each core dimension as follows:

1. Population (P): Human participants of any age undergoing job interview training, comprising healthy or typical populations such as college students and job seekers, as well as special populations with autism spectrum disorder.
2. Intervention (I): Immersive VR job interview training systems driven by AI technologies, including rule-based systems, machine learning, GenAI and LLMs.
3. Comparators (C): Studies with or without control groups, evaluating traditional face-to-face training, desktop-based training, baseline control groups, or alternative algorithmic designs within VR.
4. Outcomes (O): Studies must report at least one quantitative or qualitative evaluation metric, such as user experience, training performance, external behavioral tracking, or multimodal physiological signals (e.g., heart rate and eye movements).
5. Study Designs (S): Peer-reviewed empirical studies or experimental trials, excluding pure technical reports, reviews, or surveys.

2.3 Inclusion and exclusion criteria

To ensure the relevance and rigor of this review, a set of explicit inclusion and exclusion criteria was defined for the literature selection in strict accordance with the PICOS framework outlined in Section 2.2. These criteria operationalize the framework's broad dimensions into specific screening thresholds to capture studies investigating AI-driven interview training systems within VR environments.

Studies were included based on five primary criteria. First, existing studies confirmed that immersive VR, exemplified by HMDs, is superior to non-immersive formats (e.g., desktop VR) in eliciting psychological stress and presence comparable to real-life interviews, which is essential for exposure-therapy-based training (Martodihardjo, 2016). Therefore, to be included, the research must be conducted within an immersive VR environment, unless it initially employed a desktop VR platform whose system functionality has been fully migrated to a functionally equivalent immersive VR version (see Section 3.1, item (1) for the detailed justification regarding Migovich et al., 2021; Adiani et al., 2022; Adiani et al., 2023). Second, the focus of the study must be job interview training or preparation. Third, the job interview training system must incorporate the application of AI technology, including rule-based systems, machine learning, and GenAI. Fourth, only peer-reviewed journal articles or conference papers written in English were considered. Finally, to demonstrate the development of VR and AI technology, the review was limited to studies published in the last decade (from 2015 to 2025), based on the technical and historical baselines detailed in Section 2.1.

Studies were excluded according to several key criteria to further refine the scope and maintain a high level of technical specificity. First, studies using non-immersive VR platforms (e.g., desktop VR or mobile VR) were excluded, unless they met the exception outlined in the inclusion criteria. Second, studies were excluded if they did not focus on job interview training, such as studies about public speaking, HR recruitment, medical interviews, and workplace tasks. Third, studies that did not involve the use of any algorithms, tools or software with artificial intelligence were also excluded. Fourth, non-original research articles (e.g., review or survey papers) were excluded, as this review focuses exclusively on primary empirical studies. Lastly, publications that were fewer than four pages or failed to report key evaluation metrics, such as user experience and training performance, were excluded to maintain the review quality. These refined filters effectively narrow the scope to empirical research that aligns with the core technological objectives of this systematic review.

2.4 Quality appraisal criteria

To ensure the academic rigor and evidentiary strength of the synthesized findings, methodological quality was systematically

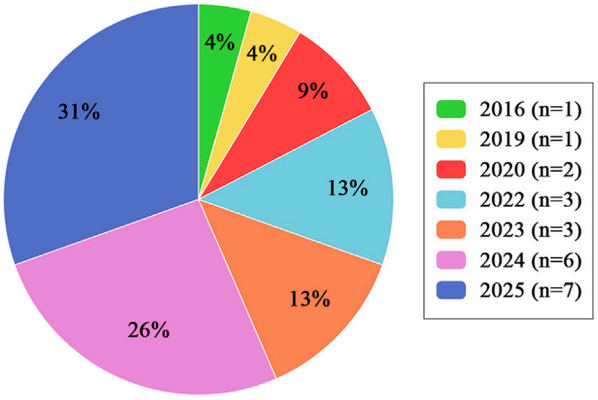


FIGURE 1
PRISMA flow diagram detailing the study selection process for the systematic review.

appraised using the Mixed Methods Appraisal Tool (MMAT) version 2018 (Hong et al., 2018). The MMAT was selected for its validated framework in evaluating diverse study designs, ranging from system-focused technical reports to user-centered empirical trials, which are prevalent in the intersection of VR and AI.

The appraisal process followed a dual-stage protocol. Each article was first screened for two foundational requirements: the clarity of the research questions and the appropriateness of the data sources. Subsequently, five category-specific criteria (Q1–Q5) were applied based on the article's primary methodology (e.g., quantitative non-randomized or quantitative descriptive). To interpret these ratings, this review strictly adhered to the MMAT guidelines, which explicitly state that the calculation of a cumulative overall score is discouraged. Instead, the guidelines advise presenting a detailed profile of each individual criterion to better evaluate the quality of the included studies (Hong et al., 2018). Following this guidance, this review adopted a descriptive, dual-reviewer evaluative approach to maximize appraisal transparency and ground the subsequent synthesis.

Under this protocol, articles were classified as High Quality if both reviewers independently verified that all five core methodological criteria were fully satisfied. Conversely, articles were categorized as Low Quality if both reviewers mutually agreed that the study exhibited systemic empirical flaws, only meeting two or fewer criteria. Remaining studies were resolved through consensus discussion between both authors until a unanimous decision was reached. Through this discussion, articles were definitively classified as High Quality if both reviewers mutually resolved the discrepancy and confirmed that all criteria were fully satisfied, allowing for standard domain-specific limitations. Alternatively, studies were assigned to Low Quality if unresolvable methodological deficiencies were established. Studies that demonstrated strong technical frameworks but exhibited specific empirical constraints like reporting omissions were synthesized as Moderate Quality.

3 Results

3.1 Literature selection and quality results

Through a systematic search, screening, and quality appraisal process, a total of 29 reports were ultimately identified for full analysis, which mapped precisely to 23 unique studies. These articles were retrieved from major academic databases, including IEEE Xplore, ACM Digital Library, Web of Science, and Google Scholar. Figure 1 illustrates the detailed process through a PRISMA flow diagram.

Initially, the database screening yielded a total of 31 reports. The methodological quality of all articles was systematically evaluated using the MMAT. To verify the reliability of this dual-reviewer assessment, the inter-rater agreement between the two independent reviewers was calculated, yielding a robust Cohen's kappa coefficient of 0.7243, indicating substantial initial consensus. All initial scoring discrepancies were comprehensively discussed and resolved between the reviewers until a 100% unanimous decision was achieved in accordance with the consensus protocol described in Section 2.4.

For maximum clarity and transparency, Table 1 presents the comprehensive evaluation matrix across all five MMAT criteria (Q1–Q5) alongside the final quality ratings, using filled and empty circles to visually denote satisfied and unsatisfied criteria. Overall, 27 articles were rated as High Quality, and 4 articles were classified as either Moderate or Low Quality. Specifically, two articles (Tian et al., 2019; Anggoro, 2020) were rated as Moderate Quality due to restricted sample sizes or incomplete reporting of evaluation metrics. Another two articles (Stănică and Moldoveanu, 2017; Stănică et al., 2018) were rated as Low Quality due to the absence of representative user testing and inferential statistics. To ensure high evidentiary strength, these two low-quality reports were subsequently excluded, leaving a total of 29 high- and moderate-quality reports for the final analysis. By excluding low-quality papers while retaining moderate-quality ones in the synthesis and clearly categorizing their limitations, this review preserves the evolutionary narrative of VR interview systems from simple rule-based scripts to advanced generative AI, prioritizing high-quality evidence to inform practical conclusions.

Following this, the remaining 29 reports were grouped to define their respective distinct research units. Aside from the 20 independent articles which each represent a single study, several research teams contributed multiple publications from the same long-term projects and were therefore consolidated as single studies:

1. The team from Vanderbilt University published five reports focusing on the Career Interview Readiness in Virtual Reality (CIRVR) system (Breen et al., 2021; Migovich et al., 2021; Adiani et al., 2022; Adiani et al., 2023; Adiani et al., 2024), which were consolidated as a single study covering its development, module validation, and evaluation. Three of these reports (Migovich et al., 2021; Adiani et al., 2022; Adiani et al., 2023) were based on a desktop VR version, as the autistic participants involved in the study showed limited tolerance for head-mounted displays. The functionally equivalent immersive VR version was developed alongside the desktop version, ensuring that the desktop-based findings remain relevant to immersive VR and thus satisfy the inclusion criteria.

TABLE 1 Methodological quality appraisal of included articles based on MMAT.

Type	Study	S1	S2	Q1	Q2	Q3	Q4	Q5	Quality
Qualitative	Dawson et al. (2022)	Y	Y	●	●	●	○	●	High
	Li et al. (2025)	Y	Y	●	●	●	●	●	High
Quantitative non-randomized	Shimizu et al. (2019)	Y	Y	●	●	●	○	●	High
	Migovich et al. (2021)	Y	Y	●	●	●	●	●	High
	Artiran et al. (2022)	Y	Y	●	●	●	●	●	High
	Artiran et al. (2024)	Y	Y	●	●	●	●	●	High
	Geng et al. (2024)	Y	Y	●	●	●	●	●	High
	Luo et al. (2024)	Y	Y	●	●	●	●	●	High
	Ueda et al. (2024)	Y	Y	●	●	●	●	●	High
	Chen et al. (2025)	Y	Y	●	●	●	○	●	High
Quantitative descriptive	Ahmed et al. (2020)	Y	Y	●	○	●	●	●	High
	Adiani et al. (2023)	Y	Y	●	●	●	●	●	High
	Ohba et al. (2022)	Y	Y	●	○	●	●	●	High
	Li et al. (2023)	Y	Y	●	●	●	●	●	High
	Si et al. (2023)	Y	Y	●	○	●	●	●	High
	Amarasena et al. (2024)	Y	Y	●	○	●	●	●	High
	Lee et al. (2024)	Y	Y	●	○	●	●	●	High
	Taheri and Tan (2024)	Y	Y	●	○	●	●	●	High
	Nofal et al. (2025)	Y	Y	●	●	●	●	●	High
	Gupta (2025)	Y	Y	●	●	●	●	●	High
	De Oliveira et al. (2025)	Y	Y	●	●	●	●	●	High
	Tasleem & Qamar (2025)	Y	Y	●	●	●	●	●	High
	Tian et al. (2019)	Y	Y	●	○	●	●	●	Moderate
	Anggoro (2020)	Y	Y	●	○	●	●	○	Moderate
	Stănică and Moldoveanu, (2017)	Y	Y	●	○	●	○	○	Low
	Stănică et al. (2018)	Y	Y	●	○	●	○	○	Low
Mixed methods	Bozgeyikli et al. (2016)	Y	Y	●	●	●	●	●	High
	Breen et al. (2021)	Y	Y	●	●	●	●	●	High
	Adiani et al. (2022)	Y	Y	●	●	●	○	●	High
	Garcia-Pi et al. (2023)	Y	Y	●	○	●	●	●	High
	Adiani et al. (2024)	Y	Y	●	●	●	●	●	High

Y: yes; N: No. Symbols: ●: Met; ○: Not met.
Questions: S1–S2: screening questions; Q1–Q5: detailed quality appraisal criteria for specific study designs.

2. The team in Japan contributed two reports centered on their interview project (Tian et al., 2019; Ohba et al., 2022), documenting the evolution from initial eye-tracking analysis to multimodal assessment. These were therefore treated as a single study.
3. The Western Washington University team contributed two reports on the InterViewR system (Ahmed et al., 2020; Dawson et al., 2022), reflecting its development from prototype testing to enhanced evaluation. They were considered as a single study.

Through this multi-stage refinement, the final pool of 29 reports mapped precisely to 23 unique studies. For consolidated studies, the publication year of the most comprehensive or primary article was used for the statistics. Figure 2 shows the distribution of included studies by publication year. Research interest in this field has grown

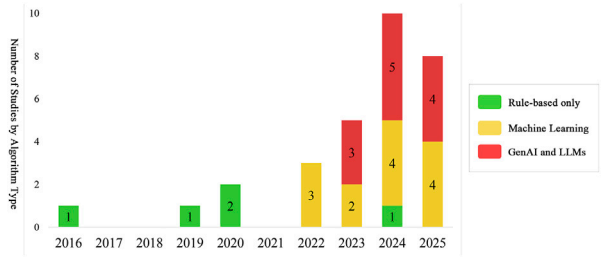


FIGURE 2
Included studies categorized by publication year.

notably since 2022, with more than half of the included studies published in the last 3 years. Figure 3 presents the year's distribution of AI techniques employed in the included studies. Studies are categorized by their primary Algorithm types. Studies where machine learning and generative AI serve as co-primary contributions are counted in both categories. Rule-based systems are only counted when they constitute the study's main technical approach. As illustrated, algorithms have evolved from rule-based systems in early years to the widespread adoption of machine learning, and recently to the emergence of large language models. This evolutionary trend provides a quantitative backdrop for the qualitative analysis presented in the following sections, while also underscoring the timeliness and necessity of conducting this systematic review.

3.2 Conceptual framework

This systematic review is structured around a logical progression encompassing VR hardware interaction, multimodal data acquisition, and the evolution of AI algorithms. This structure reflects the technical essence of the interview training system. By leveraging the high-precision data interfaces provided by increasingly mature VR hardware, these systems achieve comprehensive multimodal sensing of both the subjective inner states and objective external behaviors of participants. The core algorithms have undergone a significant evolutionary trajectory shifting from rule-based and machine learning to generative AI, thereby driving the interaction mode from mechanical responses toward context-awareness. Through this structural framework, this review constructs an analytical system integrating functional implementation, technical support, and evolutionary logic. It not only presents the capabilities of contemporary interview systems but also reveals the underlying technological evolution path.

Based on a systematic analysis of 23 independent studies, this systematic review proposes a three-layer architecture for AI-driven VR interview training systems (see Figure 4). The framework is designed to integrate functional implementation, technical support, and evolutionary logic into a unified system. Unlike conventional

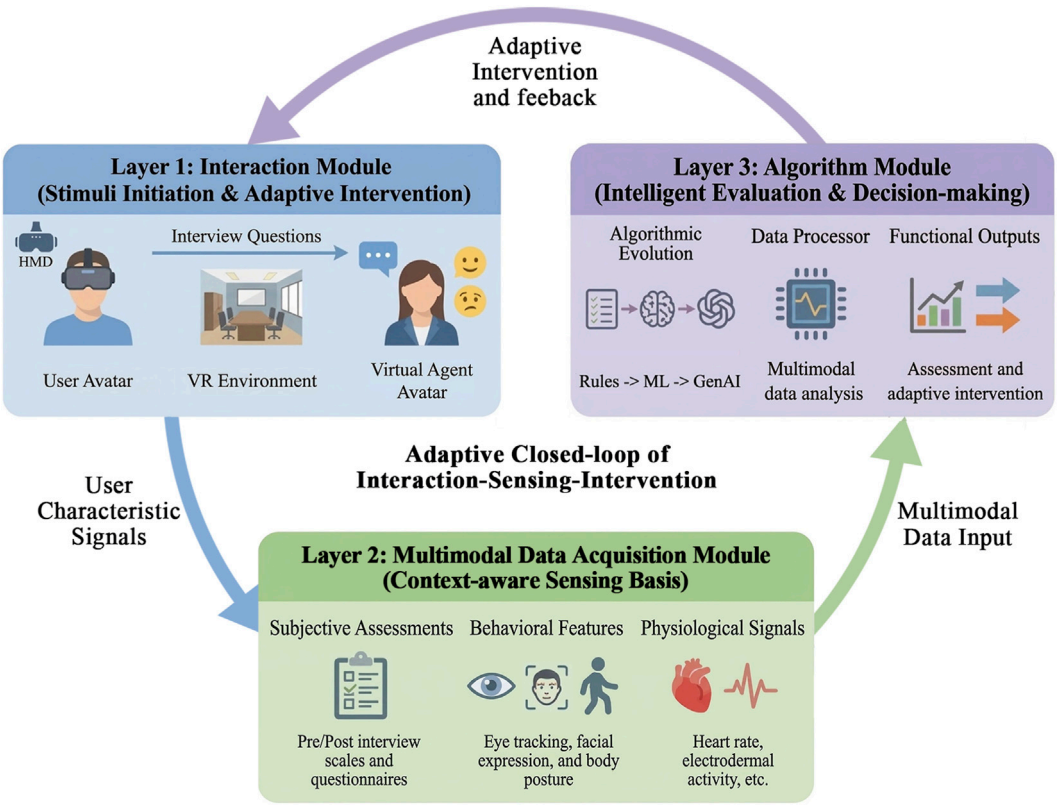


FIGURE 3
Distribution of AI algorithm types employed in the included studies by publication year.

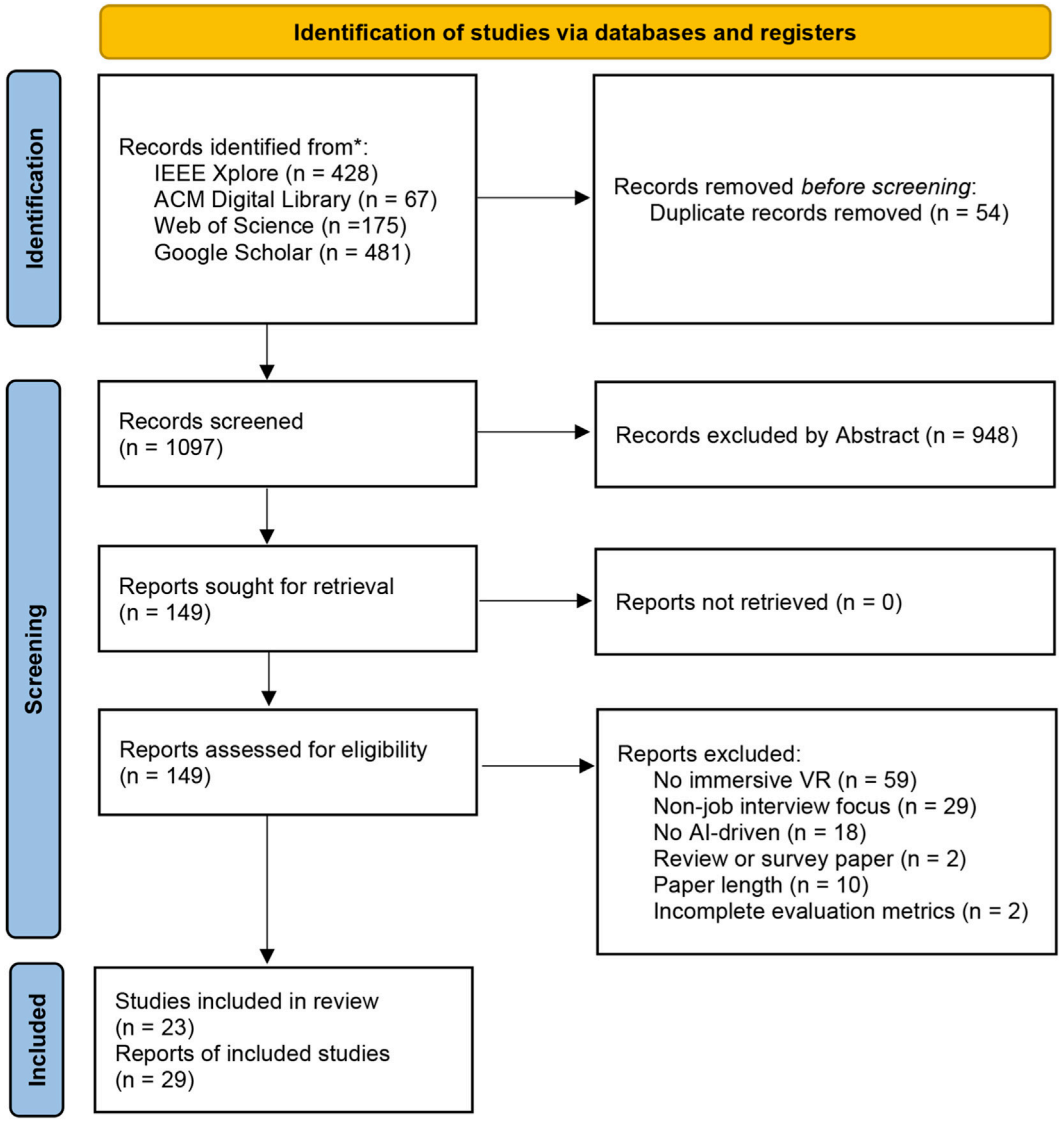


FIGURE 4
Adaptive closed-loop of interaction-sensing-intervention.

linear processes, this framework treats the system as a dynamic closed-loop, comprising the following three core modules:

1. Interaction Module: Serving as the physical and virtual interface for information exchange between the participant and the system, its core components include VR hardware (e.g., HMDs), the construction of virtual interview scenarios, and the voice-based interaction of the virtual interviewer. It provides not only the immersive environment but also the final manifestation of adaptive feedback.
2. Multimodal Data Acquisition Module: Responsible for the comprehensive capture of the participant's subjective and objective states during the interview simulation. It integrates subjective assessments including scales and questionnaires with real-time acquisition of behavioral features (e.g., facial expressions and body posture) and physiological signals such as electrodermal activity (EDA) collected by high-precision devices. This module provides the system with context-aware capabilities and establishes the data foundation and scientific evidence for subsequent algorithmic processing.
3. Algorithm Module: As the core of decision-making and evaluation, this module covers the technological evolution from early rule-based and machine learning to GenAI, while integrating a functional closed-loop of performance assessment and intervention. By analyzing data from the multimodal acquisition module, it generates adaptive functional outputs and drives the interaction module to provide real-time feedback, facilitating a fundamental shift from preset scripted logic to dynamic and adaptive interaction.

The core value of this three-layer architecture lies in constructing an adaptive intervention loop of perception, decision-making, and intervention. The system initiates initial

stimuli such as interview questions through the interaction module, captures the participant's subjective and objective multimodal data in real-time through the multimodal data acquisition module, then performs intelligent evaluation and issues intervention commands through the algorithm module, and finally guides the interaction module to execute adaptive dynamic feedback. This continuous loop ensures that the interview content remains dynamically adapted to the participant's real-time performance, thereby enhancing the authenticity and personalization of the training. Based on this architecture, the subsequent sections of this chapter will respectively elaborate on the interaction, multimodal data acquisition, and algorithm module.

3.3 Interaction module

3.3.1 VR technology and equipment

Interview training system utilizes closed-view HMDs to construct fully immersive and realistic scenarios. "Virtual Cocoon", an HMD prototype demonstrated by a British team (Garner, 2009), sought to achieve ultimate immersion by integrating multiple senses, including vision, smell, and taste. From a technological evolution perspective, hardware selection has gradually shifted from early explorations of sensory immersion toward the integration of high-precision native sensing capabilities. The most frequently employed HMDs in the selected literature are the HTC Vive series, Meta Quest series, and Fove 0. The Fove 0, released in 2016, was the first commercial headset with a native, built-in eye-tracking system, offering an accuracy of approximately 1° and a sampling rate of 120 Hz. In the study by Tian et al. (2019), the eye-tracking capability of FOVE was used to capture participants' gaze direction. By employing "collision detection" within the Unity 3D engine, the system could precisely identify the specific objects the participant was looking at. Meanwhile, HMD devices also began to evolve into more convenient forms. Oculus Go, launched in 2018, is widely regarded as a milestone product that pushed standalone HMDs into the mainstream consumer market. It features an established mobile processor and can operate independently without connecting to PC or smartphone.

Since then, high-performance HMD devices have continued to evolve. The HTC Vive Pro Eye, released in 2019, provides higher eye-tracking accuracy, ranging from 0.5° to 1.1°. Compared to FOVE 0's 90°–100° field of view (FOV), the HTC Vive Pro Eye offered a 110° FOV (Imaoka et al., 2020), as a wider FOV can deliver a stronger sense of immersion (Han et al., 2025). As shown in Figure 2, the significant increase in the number of related studies after 2020 coincided with the period of VR hardware evolution in 2019. Better hardware performance, especially smaller device sizes, lower costs, and the widespread adoption of high-precision built-in eye tracking has significantly lowered the barrier for conducting large-scale immersive VR studies. This progress also sets the stage for the AI-driven and multimodal sensing systems that will be discussed in the following sections.

Non-immersive VR, which does not require professional HMD equipment, primarily includes two common types. The first is desktop VR, which presents virtual scenes through a computer monitor. This approach bridges the accessibility gap and is particularly suitable for scenarios where devices are limited, or for individuals such as those on the autism spectrum who may not tolerate the immersive HMD experience (Breen et al., 2021; Adiani et al., 2022). The second is mobile VR, typically developed based on WebVR technology. It utilizes low-cost VR boxes like Google Cardboard paired with a smartphone, serving as a simple head-mounted virtual reality setup that delivers an immersive educational experience even on low-end smartphones. This approach is suitable for large-scale dissemination, effectively addressing the issues of high cost and large size associated with traditional HMD hardware (Anggoro, 2020). Although these two types of non-immersive VR have advantages in cost and scalability, they have fundamental technical differences compared to immersive VR. The core feature of immersive VR is six degrees of freedom (6DoF) tracking (van der Hooft et al., 2020), which enables users to move freely in virtual space. However, mobile VR only provides 3DoF for detecting head rotation and orientation, thus failing to detect the user's arm and body movements in the virtual world (Jones et al., 2025). Desktop VR, on the other hand, offers 0DoF with no spatial tracking. Their sense of presence and quality of experience are significantly lower than those of professional 6DoF devices (Dittrich et al., 2023). Therefore, mobile VR and desktop VR are typically not considered as true immersive VR.

3.3.2 VR interview scenario design

Most interview training systems have chosen Unity 3D as the integrated development platform. Its built-in XR Interaction Toolkit provides a set of pre-configured, universal interaction logic for controller input, object grabbing, UI interaction, movement, and other operations. This allows researchers to quickly construct VR experimental scenarios without writing interaction code from scratch. Additionally, Unity also has a wide range of plugins. For instance, Ueda et al. (2024) utilized the animation plugin named Final IK, which integrates four-limb tracking and facial capture data from HTC Vive series. It employs inverse kinematics to control the virtual avatar, rendering smooth and synchronized full-body movements. Furthermore, Unity's native raycast system is the core technology for research with eye-tracking. It can map the gaze direction captured by eye-tracking hardware into a 3D gaze vector ray within the virtual space in real-time. By automatically calculating collisions between this gaze vector ray and virtual objects (e.g., the interviewer's face), it enables the precise quantification of participants' fixation data (Breen et al., 2021).

Researchers have designed a diverse range of virtual environments to simulate the social settings of various industries and professions in VR interview training systems. The office and meeting room are the most common scenarios where the interviewer sits behind a desk and communicates face-to-face with the participant, aiming to simulate a formal workplace interview. To enhance the ecological validity of the training, Ahmed et al. (2020) constructed cafes and restaurants to simulate interview scenes of service industry. Targeting students, Li et al. (2025) created an indoor job fair environment to simulate common scenarios in campus recruitment, where participants needed to interact with representatives from different companies.

The core element of the interview environment is the virtual interviewer, which can display a variety of facial expressions, lip

synchronization, and illustrative gestures relevant to the conversation. The design strategy for the interviewer's appearance reflects a trade-off between stress induction and the uncanny valley effect. Through longitudinal comparison, it is observed that while realistic VR interviewers are most effective in inducing anxiety and frustration (Luo et al., 2024), an overly realistic interviewer avatar may cause discomfort due to the uncanny valley effect from its almost-but-not-quite human appearance. Consequently, to avoid interfering with the psychological test data, design strategies often prioritize an interviewer avatar with a moderate level of detail rather than extreme realism. This design indicates that the primary goal of the interview environment is not merely visual realism, but rather situational reality at a psychological level, providing a stable expectation for subsequent behavioral interactions.

3.3.3 Interaction system with virtual interviewer

The interaction with the virtual interviewer typically follows a cyclic process in the VR interview training system: (1) Speech Input, where the participant speaks their answer through the microphone; (2) Speech-to-Text (STT), where the system recognizes and transcribes the input audio into text by using speech recognition technology; (3) Response Generation, where the system determines the content of the interviewer's reply based on the participant's transcribed text through preset script, keyword matching, or LLMs; (4) Text-to-Speech (TTS), where the system synthesizes the interviewer's text-based response into audio by text-to-speech technology and playback to the participant.

Regarding STT tools, OpenAI Whisper and Google Cloud STT are widely adopted for their high recognition accuracy and great adaptation capabilities to non-ideal environments, such as diverse accents and low-quality recordings. However, current automatic recognition technologies still exhibit limitations in dealing with specific participant groups. As noted by Li et al. (2025), the system sometimes struggles to accurately identify the special speech patterns of autistic participants, such as stuttering and irregular pauses, which may cause the virtual interviewer to generate incorrect responses. To avoid errors caused by the system's automatic endpoint detection, Artiran et al. (2022) adopted a manual triggering approach where participants were required to press a button on the controller to inform the system that their response was completed. This method prevented the system from reacting prematurely due to failure to recognize speech or prolonged pauses. Although this design trade-off sacrifices some interaction naturalness, it effectively ensures the rigor of the experimental logic.

TTS enables the virtual interviewer to speak through mainstream tools including Microsoft Azure TTS, Amazon Polly, and Google Cloud TTS. Among these, Microsoft Azure TTS is recognized as one of the most advanced speech synthesis solutions, widely adopted for its high-quality neural voice technology and comprehensive support for the Speech Synthesis Markup Language (SSML). By using SSML to finely tune pitch, volume, speaking rate, and pauses, Adiani et al. (2022) developed a range of distinct vocal tones including cheerful, formal, and neutral, making the speech of virtual avatar sound highly natural and convincingly human-like. Simultaneously, the synchronization of visual performance and audio has become the key for maintaining immersion. Ueda et al. (2024) used uLipSync plugin to ensure that the virtual interviewer's lip movements were real-time synchronized with the synthesized speech. This approach prevented the disruption of the immersive experience in VR interviews caused by audio-visual desynchronization. Furthermore, although none of the studies included in this review employed it, a tool named ElevenLabs has been gaining increasing attention since 2023. Its core strengths lie in generating speech with exceptional naturalness and emotional expressiveness, alongside a voice cloning capability that can replicate a specific individual's voice from just a few seconds of audio samples. Viewed through technological evolution, speech interaction systems are transitioning from functional information transmission to high-fidelity emotion expression.

3.4 Multimodal data acquisition module

Researchers found that the actual performance of participants evaluated by experts frequently deviates from their self-reported self-efficacy, indicating that participants may overestimate or underestimate their own skill levels (Ohba et al., 2022). This discrepancy is particularly pronounced in specific populations. For instance, individuals with autism spectrum disorder usually show a lack of facial expression under stress (Lahiri et al., 2011). Similarly, introverted and shy individuals tend to suppress emotional expression under stress, deliberately controlling or concealing negative facial expressions, which makes their true emotional state difficult to detect (Geng et al., 2024). To accurately identify whether participants are being overconfident or underconfident, and to help individuals with autism or shyness to view their own unconscious emotional responses, many researchers collect participants' behaviors (e.g., eye movements, facial expressions, body gestures) and physiological data (e.g., heart rate and galvanic skin response). This method can help participants better understand their current feelings and improve self-regulation. Furthermore, collecting multimodal data can enable the interview training system to sense the participant's current state and allow for personalized training adjustments.

3.4.1 Perception and questionnaires

For participants, a positive VR interview training experience is rooted in their subjective perception of simulated social interaction. A comprehensive assessment of training effectiveness requires the use of standardized questionnaires and scales to capture these subjective experiences, including participants' level of anxiety, system usability, and the sense of immersion in the VR environment. These findings are then cross-referenced with objective multimodal behavioral data to form a comprehensive research evaluation loop. Table 2 presents an overview of the questionnaires utilized in the reviewed studies, which have been systematically categorized into the following six major groups:

1. Participant screening and trait assessment: To ensure the subject accuracy and control for individual differences in training effectiveness, researchers conduct baseline assessments like the Social Responsiveness Scale to quantify the severity of social impairment and performance in social interaction (Chen et al., 2025). This

TABLE 2 Summary of commonly used questionnaires in reviewed studies, categorized by measurement type.

Usage	Questionnaire	Application papers
Participant screening and trait assessment	Autism-spectrum quotient	Artiran et al. (2022); Chen et al. (2025)
	Social responsiveness scale	Adiani et al. (2022); Chen et al. (2025)
	Wang Qianqian's college shyness scale	Geng et al. (2024)
Anxiety and mental health (pre-/Post-interview)	Measure of anxiety in selection interviews	Luo et al. (2024)
	General self-efficacy	Ohba et al. (2022); Luo et al. (2024)
	Anxiety rating scale	Chen et al. (2025)
VR system acceptance and usability	System usability scale	Artiran et al. (2022); Tasleem and Qamar (2025)
	Technology acceptance model	Tasleem and Qamar (2025)
VR immersion and side effects	Presence questionnaire	Shimizu et al. (2019)
	Igroup presence questionnaire	Ueda et al. (2024)
	VR sensitivity scale	Ahmed et al. (2020)
	Simulator sickness questionnaire	Chen et al. (2025)
	Virtual embodiment questionnaire	Ueda et al. (2024)
Training effectiveness	Performance rating scale	Luo et al. (2024)
	Adapted MIRS-Chinese	Chen et al. (2025)
	Questionnaire for prosody, behavior, and response	Ueda et al. (2024)
Cognitive workload	NASA task load index	Luo et al. (2024)

assessment is a prerequisite for ensuring internal validity and understanding the disparities in intervention outcomes.

2. Quantification of anxiety and mental health (pre-/post-interview): As a form of Exposure Therapy, the effectiveness of interview training depends largely on its ability to induce and alleviate psychological stress. The Measure of Anxiety in Selection Interviews evaluates interview anxiety across five dimensions including communication, appearance, social, performance, and behavior (Luo et al., 2024). This assessment provides a direct psychometric basis for measuring the effectiveness of the exposure therapy.
3. Evaluation of VR system acceptance and usability: The effectiveness of interview training depends mainly on user engagement and the usability of the system. For instance, the System Usability Scale can be used to assess the usability and ease of operation of a system (Ahmed et al., 2020). Lower usability ratings often indicate that participants may experience frustration due to technical barriers, causing the loss of learning motivation.
4. Evaluation of VR immersion and side effects: Immersion is the core metric for measuring VR quality, as it is directly related to the participant's sense of presence. The Presence Questionnaire is used to measure participants' subjective involvement, immersion, and sensory depth (Martodihardjo, 2016). Simultaneously, high-intensity immersive experiences may lead to the onset of cybersickness. The Simulator Sickness Questionnaire is specifically designed to test whether participants experience adverse reactions, such as nausea, eyestrain, or dizziness while using a VR HMD (Ueda et al., 2024), ensuring that the VR environment maintains both psychological validity and physiological tolerability.
5. Evaluation of training effectiveness: Due to the current paucity of validated and widely accepted scales assessing job interview performance, researchers of reviewed literature commonly develop their own scales to evaluate the effectiveness of interventions. For instance, Chen et al. (2025) translated the Mock Interview Rating Scale (MIRS), designed by Smith et al. (2023) to evaluate the interview performance of individuals with schizophrenia, into Mandarin Chinese. They further adapted the scale by incorporating additional assessment items specifically tailored for autistic individuals and named it "adapted MIRS-Chinese". These assessments validate the performance improvements of participants in interview training and provide a scientific basis for subsequent personalized guidance by identifying specific deficiencies in their performance.
6. Cognitive workload assessment: Excessive workload can induce mental fatigue and impair performance, making it essential for interview training systems to balance task difficulty with the participants' psychological resilience. The NASA Task Load Index evaluates workload across six dimensions, including mental demand, physical demand, temporal demand, performance, effort, and frustration (Luo et al., 2024). Through continuous monitoring of the workload, systems can achieve user-centered adaptive training, allowing for the dynamic optimization of interview difficulty.

3.4.2 Non-verbal behavior acquisition

Facial expressions can not only be used to assess the emotional state of individual but can also trigger emotional responses of the observers, thereby influencing their behavior in social interactions (Frith, 2009). Based on Ekman's (1993) theory of eight basic emotions (happiness, surprise, fear, anger, disgust, contempt, neutral, sadness), Adiani et al. (2024) used traditional image capture and the Microsoft Azure Face API to generate confidence scores and determine participant's emotional state. In contrast, recent studies have shifted toward deep learning-based facial behavior analysis tools such as OpenFace. By utilizing high-precision facial landmark recognition and eye-tracking, Amarasena et al. (2024) can detect participants' emotion, attention, and engagement in real-time, marking the field's progression toward a higher level of feature extraction. In addition to visual cues, speech serves as another vital behavioral metric. Increased stress during interviews can cause changes in participants' volume and voice quality, which were captured by microphones (Ahmed et al., 2020; Dawson et al., 2022). Chen et al. (2025) not only recorded volume and pitch but also utilized openSMILE to extract more granular acoustic features such as jitter, shimmer, and harmonic-to-noise ratio to quantify the participant's anxiety levels. This depth of analysis provides high-fidelity data for precise interventions in subsequent algorithmic modules.

Eye gaze plays an important role in social interaction, serving both to perceive others' states and to convey one's own attention (Cañigueral and Hamilton, 2019). In actual recruitment, more direct eye contact is perceived as a sign of interest and increases the chances of being hired, while more frequent eye contact avoidance or eye wandering can be seen as a lack of enthusiasm and incompetence (Forbes and Jackson, 1980). These social expectations present significant challenges for neurodiverse groups, as individuals on the autism spectrum typically exhibit a characteristic avoidance of direct eye contact. To eliminate employers' biases towards individuals with autism, Breen et al. (2021) utilized the native eye-tracking of the HMD to define specific Regions of Interest (ROI) on the interviewer's eyes and face, as well as objects in the office environment like a wall clock, windows, and a cup on the desk. Gaze data was collected and quantified through heat maps and scatter plots to quantify gaze duration and frequency. This approach helps employers understand that the gaze patterns of individuals on the autism spectrum have no correlation with their professional competence or focus, reflecting an evolution of VR training systems toward supporting inclusive employment.

Embodied Cognition theory posits that there is a bidirectional influence between body posture and psychological state (Niedenthal, 2007). During an interview, participants often exhibit unconscious physical behaviors due to stress or anxiety, which can easily leave a negative impression on the interviewer (Feiler and Powell, 2016). Geng et al. (2024) employed the Microsoft Kinect Body Basics to identify and track 25 key skeletal joints of the human body, including the head, neck, shoulders, hands, wrists, etc. This enabled the system to automatically detect improper behaviors ranging from crossing arms, body shaking, fidgeting, or playing with hair. Furthermore, the potential of acquired behavioral data extends beyond real-time detection. Switching from an observer's perspective allows individuals to review their own performance as an audience, which can help them form a more objective self-assessment, especially for those who lack confidence to express themselves (Zhou et al., 2021).

3.4.3 Physiological signal measurement

Behavioral signals provide participants with a mirror for self-observation. However, physiological reactions triggered by high-intensity stress during an interview (e.g., sweating and accelerated heart rate) are regulated by the autonomic nervous system and are not consciously controlled (Migovich et al., 2021). Therefore, to obtain more objective and in-depth evidence, researchers employ physiological signal acquisition technology for measurement and supplementation. The Empatica E4 wristband can collect participants' inter-beat interval, blood volume pulse, as well as skin conductance level and skin conductance response through EDA measurements. The Empatica E4 wristband is widely used for stress and emotion assessment as its data measuring quality and wearability balancing. The device measures skin conductance by applying a minimal current between two skin-contact electrodes. When a participant experiences emotional fluctuations, increased cognitive load, or stress, excitation of the sympathetic nervous system can cause changes in sweat gland activity, thereby resulting in variations in skin conductance (Luo et al., 2024). To accurately predict participants' self-efficacy, Ohba et al. (2022) built a prediction model by collecting four types of data including acoustic, physiological, eye-tracking, and linguistic to check whether they were being overconfident or underconfident. The experimental results demonstrated that physiological features captured by the E4 wristband achieved the highest accuracy in predicting self-efficacy. This further confirms that self-efficacy, as an internal and subjective self-perception, exhibits a stronger correlation with involuntary physiological responses that are difficult to consciously control, such as heart rate fluctuations and changes in EDA.

A lightweight wearable device known as the Hexoskin smart shirt offers higher precision and quality in cardiorespiratory data acquisition compared to the E4 wristband. Dawson et al. (2022) utilized the Hexoskin smart shirt to focus on capturing participants' R-R intervals, heart rate, and breathing rates for real-time stress level monitoring. This physiological assistance enables both participants and job coaches to identify specific interview questions that elicit the highest level of physiological stress, thereby facilitating more targeted and effective training. However, the authors also noted that the major drawback of Hexoskin lies in its high-cost barrier to adoption, which conflicts with the development team's original intention of enhancing the accessibility of the interview training system through affordable hardware. Furthermore, NeuroSky (2009) utilized single-lead dry electrode technology, and its captured Electroencephalogram (EEG) signals have been proven to be highly similar to those collected by the professional benchmark device, BIOPAC. Geng et al. (2024) employed the NeuroSky EEG headband to analyze participants' brainwaves, allowing the system to calculate their attention and meditation levels as a basis for the virtual interviewer to adjust its attitude in real-time. The establishment of such physiological feedback loops signifies

a shift from unidirectional recording toward dynamic closed-loop interaction within the multimodal data acquisition module.

3.5 AI-algorithms-driven decision and evaluation modules

3.5.1 Rule-based methods

Early virtual interview systems primarily relied on rule-based methods. These systems were driven by preset decision trees and "if-then" rules to ensure that the interview interaction process remained highly structured and predictable. In terms of behavioral control, the logical transitions and systemic states of the virtual interviewer are driven entirely by explicit state transitions. Tian et al. (2019) utilized state machines to drive the behavioral logic of the virtual interviewer, automatically switching between questioning and listening states. Notably, researchers have attempted to enhance the system's dynamism and interactivity through keyword extraction. For example, the Interactive Question Generator system designed by Shimizu et al. (2019) extracts keywords (e.g., school work or programming) from the participant's responses and uses them to generate follow-up questions (e.g., "What did you learn from school work?"). While this design allows the virtual interviewer to appear as though they are listening attentively, its essence remains a logical transition within limited branches, making it difficult to respond flexibly to the participant's complex linguistic feedback.

The assessment and feedback logic of the rule-based systems are also standardized and deterministic. Geng et al. (2024) employed strict geometric rules to identify improper postures. For instance, the system calculates the distance between the participant's hand joints and the head. If the distance is less than 8 cm and remains for more than 2 s, it is judged as scratching the head. The system then provides immediate voice feedback to help participants consciously control these unconscious nervous behaviors. Regarding interview performance evaluation, a performance score ranging from 0 to 100 was automatically calculated, based on completion time, the number of prompts given, and the number of incorrect actions during the interview, which was then provided to the participants after the interview (Bozgeyikli et al., 2016). However, rule-based mode relies heavily on pre-programmed rules and lacks the capacity for semantic understanding and generative dialogue. The virtual interviewer cannot comprehend the specific content of the answers, making the training a one-way questioning session. The interaction experience is frequently criticized as "mechanical" and "stiff" by participants (Menon, 2021), thereby prompting subsequent research to explore machine learning algorithms with automated feature-learning capabilities.

3.5.2 Machine learning

As the demands for intelligence, personalization, and the depth of natural interaction in interview training systems continue to rise, the limitations of rule-based systems are becoming increasingly apparent. Therefore, AI algorithms capable of semantic understanding, learning interaction patterns, and generating adaptive feedback have become a key evolution direction for virtual interview systems. Machine learning utilizes algorithms to automatically learn from historical and multimodal data, building intelligent models capable of generating adaptive dialogue and predicting interview performance. As summarized in Table 3, we categorize machine learning techniques utilized across the selected studies by their specific algorithm names, detailing their application purpose, strengths, and weaknesses within specific application contexts. It is important to note that when summarizing the machine learning algorithms for the 23 included unique studies, certain long-term research projects (e.g., CIRVR system) consolidated from multiple publications, where different articles may focus on distinct algorithmic modules to address specific research challenges. Consequently, these algorithms are listed individually in the table to accurately reflect the technical characteristics. For instance, Adiani et al. (2024) utilized the strengths of Random Forest in handling non-linear data and assessing feature importance to develop automated stress detection models. The study by Ohba et al. (2022) utilized Long Short-Term Memory (LSTM) to process time-series data from the question-answer sequences in interviews, combining multimodal data captured by the Empatica E4 wristband to train regression models for predicting the interview performance of participants. Additionally, machine learning has demonstrated unique value in enhancing perceptual precision. Artiran et al. (2022) introduced Kalman filtering and spatio-temporal density-based clustering to eliminate interference from blinks and gaze jitter, enabling the precise quantification of inherent differences in gaze behavior between individuals on the autism spectrum and neurotypical individuals.

Machine learning empowers the system with data-driven predictive assessment and real-time adaptive feedback. By automatically processing collected multimodal data including facial expressions, speech, physiology, and eye gaze, participants and job coaches can intuitively view various data visualizations, such as fixation scatter charts, gaze heat maps, and conversation sunburst graphs, to formulate targeted training protocols (Adiani et al., 2024). Some training systems provide a variety of real-time feedback prompts. For instance, specific icons may emerge within the virtual interview environment: a red heart icon indicated that the participant's stress level was too high, a speaker icon signified that the response volume was too low or too high, and an eye icon denoted a break in eye contact (Ahmed et al., 2020). More importantly, machine learning has helped the VR interview training system to achieve, for the first time, a complete real-time closed loop of perception, decision-making, and intervention. For example, Migovich et al. (2021) developed a stress detection model based on physiological data collected by Empatica E4 wristbands, combined with K-nearest neighbor classifier. When a participant was detected in a high-stress state, the virtual interviewer would automatically switch to simpler questions to prevent frustration arising from excessive anxiety. Once the participant's emotions stabilized, the system would return to more challenging questions. To address the scarcity of online interview data, Li et al. (2023) utilized pre-trained Bidirectional Encoder Representations from Transformers (BERT) to propose a disentangling mechanism to pre-train a knowledge selector and a

TABLE 3 Technical comparison of machine learning algorithms.

Machine learning	Application purpose	Strengths	Weaknesses	Source
Convolutional neural network	Identifying non-verbal cues (e.g., facial expressions, body language)	Superior spatial feature extraction; real-time micro-expression analysis	Complex development and training; highly dependent on large-scale datasets	Adiani et al. (2022); Amarasena et al. (2024); Gupta (2025)
Key-value memory network	Resume knowledge extraction, personalized question generation	Handles domain-specific technical terms; effective in low-resource scenarios	Sensitive to knowledge noise; requires extensive pre-training	Li et al. (2023)
Kalman filtering	Smoothing 3D gaze data to improve eye-tracking precision	Smooths real-time signals; corrects for VR headset slippage or blinks	Limited effectiveness with extreme non-linear gaze jumps	Artiran et al. (2022)
Spatio-temporal density-based clustering	Fixation identification and noise filtering	No need to pre-define cluster number; handles temporal consistency	Clustering effectiveness dependent on hyperparameter tuning	Artiran et al. (2022)
K-nearest neighbor	Physiological stress detection and emotional state classification	Good at processing physiological data with high classification accuracy	Performance depends heavily on precise tuning of hyperparameters (e.g., K-value)	Migovich et al. (2021)
K-means	Skill identification and classification from resumes	Provides finer granularity in skill clustering	Complex preprocessing; difficulty in data cleaning and optimal K-value	Amarasena et al. (2024)
Random forest	Analysis of physiological data and dialogue act classification	Highly effective in physiological signal processing; offers good predictive stability	Lacks deep semantic understanding; unsuitable for complex text classification	Breen et al. (2021); Adiani et al. (2022); Adiani et al. (2024)
Sentence-BERT	Coherence analysis of interview responses	Generate semantic vectors; excels at semantic similarity computation	Average performance in complex reasoning or real-time generation	Chen et al. (2025)
Bidirectional and auto-regressive transformers	Assess robustness, analyzing semantic similarity of questions by different prompts	Detects stability and anti-interference; features strong semantic understanding	Potential biases (e.g., religion); content may be overly homogenized or generic	Nofal et al. (2025)
LSTM and bidirectional LSTM	Time-series processing; dialogue context and interview performance prediction	Robust sequential modeling; captures long-range dependencies	Limited in predicting complex socio-psychological traits	Ohba et al. (2022); Gupta (2025)
GenAI and LLMs	Content generation; dialogue and behavior generation; visual generation	Flexibility and adaptability; context awareness; reduced need for manual scripting	Algorithmic bias and hallucinations; ethical and privacy risks; limited controllability of dialogue	Si et al. (2023); Amarasena et al. (2024); Lee et al. (2024); Luo et al. (2024); Li et al. (2025); Nofal et al. (2025)
Robustly optimized BERT pretraining approach	Sentiment analysis and standardized performance quantification	High accuracy; standardized output	Risk of algorithmic bias (e.g., nationality, religion, gender)	Nofal et al. (2025)
Regression analysis models	Performance prediction, causal inference, and feature correlation	Powerful quantitative interpretability; clear importance weight analysis	Dependent on feature engineering; average time-series data capability	Tian et al. (2019); Ohba et al. (2022); Luo et al. (2024)

dialogue generator separately. This approach enables the system to comprehensively understand resumes and job descriptions, subsequently generating highly relevant and professional mock interview questions. Collectively, these studies signify an intelligent transformation of virtual interview training systems toward a future capable of contextual understanding, stress sensing, and human-like predictive analysis.

3.5.3 GenAI and LLMs

GenAI aims to learn distributions of existing data to create entirely new content, such as text, images, and audio, and has demonstrated significant application potential across various fields in recent years (Si et al., 2023). Since the rise of LLMs in late 2022, their powerful context-awareness and dynamic dialogue

generation capabilities have rapidly made them a technological focal point in the AI field, while also offering a new developmental path for intelligent interview training systems. Simultaneously, models specializing in visual content synthesis have also achieved breakthroughs. Although GenAI remains within the technical framework of machine learning, its uniqueness in terms of parameter scale, versatility, and generative capacity is frequently discussed as an independent technology in research and application. This breakthrough is first manifested in the rapid construction and diversification of visual assets. For instance, Si et al. (2023) used Stable Diffusion to generate highly realistic 2D virtual interviewer images and integrated them into the VR 3D space. By adjusting prompt parameters, they rapidly produced diverse and realistic interviewer avatars, providing a novel approach to the visual construction of VR interview scenarios and addressing the limitations of traditional VR environments where virtual human assets often lack variety and realism.

As the core of GenAI in semantic comprehension and text generation, LLMs have broken through the limitations of traditional rule-based or fixed dialogues, enabling the dynamic generation of personalized natural language feedback based on conversational context. The deep integration of LLMs with virtual reality allows virtual interviewers to generate highly adaptive and realistic dialogues, thereby constructing a more immersive and emotionally resonant interview training experience (Li et al., 2024). This evolution significantly enhances the social presence and autonomous decision-making capabilities of virtual characters. Specifically, Li et al. (2025) utilized LLMs and prompt engineering to simulate unpredictable interview scenarios, such as a virtual interviewer with aggressive tone, to test participants' resilience under situational social stress. This autonomous dynamic interactivity generated by LLMs imposes realistic psychological pressure on participants, which is unattainable by previous rule-based systems. Furthermore, Amarasena et al. (2024) utilized the T5 model to generate personalized interview questions based on participants' uploaded resumes and intended job positions, while providing text-based grammatical corrections and response suggestions based on the participants' answers. This approach demonstrates the professional potential of LLMs as intelligent educational scaffolds.

The superior semantic understanding of LLMs imparts high linguistic flexibility to the system, which not only enhances scenario adaptability but also effectively breaks the limitations of interview training in cross-cultural applications. Nofal et al. (2025) utilized ChatGPT to generate dynamic interview questions and provide personalized feedback based on students' responses, developing the Interview Training and Education Module (ITEM) within a metaverse environment. LLMs possess critical value regarding linguistic adaptability, allowing training systems to flexibly serve multilingual and diverse scenarios rather than being restricted to a single cultural or linguistic framework. In a specific regional context, Lee et al. (2024) leveraged ChatGPT to develop a chatbot supporting Cantonese communication, providing interactive interview exercises that incorporate English, Mandarin, and Cantonese to adapt to the unique requirements of the Hong Kong workplace environment. However, the powerful generative ability of LLMs is inevitably accompanied by new challenges. Many studies have reported risks such as AI hallucinations, content bias, and uncontrollable dialogue, which are discussed in detail in Section 3.7.

3.6 Research participants and training objectives

Based on a systematic review of the literature, the targeted research participants range from specific groups facing significant employment barriers to a broad spectrum of general job-seeking graduates and vocational guidance personnel. Researchers placed a deep focus on the autism spectrum population, including autistic adults (Breen et al., 2021; Adiani et al., 2024), university students (Artiran et al., 2024), and high-functioning autistic individuals who possess high cognitive abilities but encounter social communication challenges (Bozgeyikli et al., 2016). For interview training targeted at individuals with intellectual disabilities, Garcia-Pi et al. (2023) specifically reduced the speaking speed, conversation length, and linguistic depth of the virtual interviewer to better meet their specific communication needs and learning characteristics. Furthermore, De Oliveira et al. (2025) focused on assisting pre-released individuals with social re-entry training, helping them practice answering sensitive questions regarding their past backgrounds. By mastering these high-stakes social interactions in a virtual setting, participants can rebuild the necessary self-confidence and professional identity required for a successful transition back to the modern workforce.

At the same time, academic studies extensively covered technical interview training for students majoring in computer science and software engineering who need to demonstrate both coding skills and soft skills (Tian et al., 2019; Amarasena et al., 2024; Lee et al., 2024). Beyond technical fields, simulated interview dialogue practice for non-native English speakers (Anggoro, 2020), and interview skill enhancement for current students, recent graduates, and general young job seekers (Ohba et al., 2022; Si et al., 2023; Geng et al., 2024; Luo et al., 2024; Taheri and Tan, 2024; Ueda et al., 2024). For instance, Geng et al. (2024) implemented gradient training specifically for college students with high shyness, which led to a significant reduction in participants' body language anxiety and average heart rate, alongside improvements in the linguistic coherence and logical fluency of their responses.

Additionally, participants require continuous monitoring by job coaches when facing challenging scenarios to prevent high-pressure environments from causing psychological distress or trauma to autistic individuals. In practice, the presence of a human supervisor allows for immediate intervention if a participant shows signs of extreme anxiety during the VR simulation. Therefore, as key facilitators of the interview training system, job coaches (Li et al., 2025), career counselors (Adiani et al., 2024), and corporate employers as well as human resources specialists (Breen et al., 2021) have all been incorporated into the research scope to assist in training, evaluate system utility, and construct support networks. The high diversity among participants underscores the core value of intelligent VR interview training systems in promoting equitable employment and inclusive development.

3.7 Assessment validity and ethical considerations

Standardized questionnaires may exhibit issues such as perceptual disparities, insufficient applicability, and algorithmic biases throughout their design, application, and interpretation, particularly when these tools are applied to specific groups or populations from diverse cultural backgrounds. First, cultural differences can lead to systematic perceptual biases. Individuals from East Asian backgrounds, particularly those from Japan, tend to underestimate their own skill levels, which can result in lower self-efficacy scores reflected in questionnaire results (Heine and Hamamura, 2007; Heine et al., 2000). Secondly, questionnaires may lack sufficient measurement validity for specific groups. The System Usability Scale (SUS) is used to evaluate the feasibility of interview systems, but it struggles to distinguish between "difficulties arising from operation" and "difficulties arising from social or psychological pressure." This led to autistic participants reporting a median SUS score (57.5) that was significantly lower than the score given by neurotypical participants (77.5) (Adiani et al., 2022). This demonstrates that standardized questionnaires are not objective measures. In interpreting their results, researchers must take full account of their limitations in cross-cultural and cross-group contexts and avoid equating measurement outcomes with absolute truth.

With the refined acquisition of multimodal data, the protection of participant privacy and security has gained increasing attention. Current studies emphasize a dual approach, combining physical storage isolation with logical access control to safeguard data sovereignty. All participants' identity records must be de-identified, and personal identifying information must be completely removed (Li et al., 2023). Participants were only permitted to log in and view feedback from their own interviews to prevent security breaches between different participants (Dawson et al., 2022). Breen et al. (2021) noted that all captured physiological and eye-tracking data were stored through randomly generated Participant IDs, ensuring the data remains de-identified within the SQL database. Additionally, all recording data should be ensured to have any identifiable information removed and will be used only for internal research purposes, without being shared with any external parties (Nofal et al., 2025).

While LLMs have significantly enhanced the depth of system interaction, they may produce biased outputs, which can lead to unfair evaluation scores and discriminatory feedback in the interview training systems. Such outcomes hinder the confidence and growth of learners, directly contradicting the fundamental goals of equitable education and inclusive workforce development. By artificially manipulating participants' backgrounds, including countries of origin, religion, and gender, while keeping their interview responses completely identical, Nofal et al. (2025) confirmed that LLMs used as scorers indeed exhibit biases. Among the models tested, ChatGPT proved to be the most sensitive to the background variations, assigning the lowest scores to UK males and Buddhists from Zambia. Meanwhile, BlenderBot gave the lowest scores for participants associated with Qatar, Angola, and Zambia. Additionally, LLMs may exhibit issues such as ethical inconsistencies, unreliable content, and deviations from expectations when generating interview content. ChatGPT still faces obstacles in comprehending complex concepts and domain-specific knowledge. Such challenges are often linked to the way instructions are formulated. For instance, when writing prompts, job coaches tend to rely on human-to-human instructional experiences or over-generalization, which leads LLMs to accurately understand their instructions (Zamfirescu-Pereira et al., 2023). However, when coaches incorporated more specific details into their prompts, LLMs demonstrated significant performance improvements (Li et al., 2025).

Fundamentally, the bias and reliability issues of LLMs arise from three factors. First is the bias within training data, ChatGPT's training data over-represents youth populations, developed nations, and English speakers. Second, due to the dynamic nature of LLMs, even with meticulously designed prompts, the conversational content of virtual interviews may still move into unexpected directions (Li et al., 2024). Third, its generated content prioritizes anthropomorphism over accuracy, tending to produce lengthy responses that lack verifiable sources of truth (Nofal et al., 2025). Consequently, when integrating LLMs into interview training systems, cautious measures should be implemented, such as providing non-expert users with pre-set text templates, establishing human review and content correction mechanisms. To address the forgetfulness issue of LLMs in long-duration interview conversations, a multi-layered memory mechanism (e.g., background memory, initial memory, and latest memory) can be designed to ensure that LLMs identify the uniqueness of each participant and maintain the long-term conversation process (Garcia-Pi et al., 2023). Researchers and job coaches must clearly inform participants that the virtual interviewers' responses are generated by AI rather than a real human. Furthermore, they should assist participants in handling inappropriate language or challenging scenarios to ensure that ethical standards are strictly upheld throughout the training process.

4 Discussion

This chapter, built upon the preceding detailed review of VR interview training system architectures and technical modules, will focus on exploring and addressing the three research questions proposed in Section 1. These questions include: RQ1, How does AI drive VR interview training systems to evolve from early preset scripts to context-awareness and adaptive interaction in the current literature; RQ2, How does the current literature describe the multimodal data fusion used to achieve a comprehensive assessment of participants' external performance and internal states; RQ3, What are the challenges identified in the current literature for AI-driven VR interview training systems.

4.1 RQ1: technology evolution from pre-scripted to adaptive interaction

The intelligence process of VR interview training systems is an evolution of interaction from mechanical response to adaptive perception. Driven by the iteration of AI algorithms, this evolution follows a three-stage path that progressively integrates Interaction, Multimodal Data Acquisition, and Algorithm modules into an increasingly complex, user-centric feedback system.

The early stage was dominated by rule-based systems, where the interaction loop was highly rigid. In these systems, the multimodal data acquisition layer was limited to simple inputs, typically relying on pre-defined keyword matching. For instance, early studies utilized Artificial Intelligence Markup Language to configure a series of questions and answers, where the system determined the correctness of technical responses by checking for specific keywords (Stănică and Moldoveanu, 2017; 2018). The algorithm layer utilized state machines to execute fixed logical branches, which in turn drove the interaction layer to produce predictable and mechanical responses from virtual interviewers. The inability of these algorithms to perform specific semantic recognition or interpret non-verbal cues resulted in a lack of flexibility to adapt to the user's real-time performance.

Machine learning marks a leap toward data-driven, predictive interview training systems. During this stage, the capabilities of the multimodal data acquisition module were significantly enhanced to capture linguistic, behavioral, and physiological signals. The algorithm module shifted from manual rules to learning from these datasets to predict interview performance or recognize emotions. This allowed for a personalized closed loop of perception, decision-making, and intervention. For instance, research by Amarasena et al. (2024) showed how capturing data on participants' emotions, attention, and engagement allows the algorithm module to generate a comprehensive confidence score, which then informed the feedback provided in the interaction module. Moreover, as a bridge transitioning toward GenAI, the system utilized a retrieval-augmented generation model to retrieve data from external knowledge bases and generate a highly personalized career development recommendation report based on the participants' aggregate interview performance.

Currently, LLMs are advancing interview systems toward generativity and deep personalization. In this stage, the algorithm module provides deep semantic reasoning, upgrading the multimodal data acquisition module from simple signal detection to a broader understanding of context. Consequently, the interaction module can generate real-time, context-aware dialogue. Specifically, these systems can parse user resumes to dynamically generate highly tailored interview questions and automatically evaluate the factual accuracy of responses for objective scoring (CJ et al., 2024). Moreover, the prompting capability of LLMs has greatly reduced the threshold for customizing professional scenarios. Job coaches can now use natural language prompts to define the interviewer's personality, environment, and practice goals, allowing for flexible generation of diverse training scenes (Li et al., 2025).

In summary, the advancement of AI technology has transformed VR interview systems from simple tools that execute pre-set programs into intelligent interviewers capable of multimodal perception, semantic understanding, and dynamic generation of personalized content.

4.2 RQ2: comprehensive assessment based on multimodal data fusion

Assessments based on a single dimension often fail to capture the dynamic complexity of interviews as social activities. Consequently, research increasingly adopts multimodal data fusion systems, utilizing the Multimodal Data Acquisition module to holistically analyze participants by cross-validating subjective perceptions, external behaviors, and internal physiological signals.

Researchers use standardized questionnaires to quantify subjective perceptions, including user traits, anxiety levels, system acceptance, VR immersion, and cognitive load. For instance, as training effectiveness depends significantly on usability and appropriate interaction, the SUS is widely used to measure technical acceptance, operational complexity, and user confidence. These subjective data are indispensable for understanding participants' psychological experience and overall intervention effectiveness.

The acquisition of behavioral signals quantifies subtle micro-data from an external level that is usually imperceptible during an interview. Eye tracking not only identifies gaze avoidance symptomatic of social anxiety but also reveals unconscious cognitive load and visual attention patterns. Real-time facial and speech analysis provides objective indicators of affective state and fluency. Moreover, motion-capture-based posture analysis can automatically detect inappropriate movements (e.g., crossed arms or legs) and alert participants to control them consciously, ensuring that interviewers maintain a professional posture while expressing themselves verbally (Jadhav et al., 2024). Ueda et al. (2024) used Vive Trackers to capture the movement of participants' limbs and integrated it with eye-tracking and facial expression capturing, allowing participants to observe their own virtual avatars' body movements and facial expressions from a third-person perspective. A recent study highlighted that transitioning to an interviewer's perspective is significantly more effective in upgrading the self-assessment of verbal and non-verbal communication skills than the traditional first-person perspective (Seinfeld et al., 2024). This approach was designed to stimulate participants' objective self-awareness, helping them to adjust their verbal and non-verbal behaviors.

The collection of physiological signals provides the deepest and most difficult-to-disguise internal evidence. The significance of physiological data lies in reactions absent from overt behavior or self-report. A participant may appear calm while experiencing a racing heart or sweating. Research shows that physiological features can surpass behavioral data in predicting states like self-efficacy (Ohba et al., 2022).

A comprehensive assessment of participants depends on the fusion of multimodal data. The integrated analysis of subjective reports, objective behavior, and physiological signals greatly enhances scientific rigor and real-time capacity, providing a comprehensive sensing basis for personalized feedback and intervention.

4.3 RQ3: challenges facing the system

Although AI-driven VR interview training systems are developing rapidly and showing great potential, they still face a series of challenges in technical efficiency, assessment validity, and ethical fairness, which restrict their reliability and universal application.

VR hardware struggles with an "immersion-convenience-cost" trilemma. Professional HMDs can provide high-precision eye

tracking and wider fields of view but sacrifice freedom and accessibility. Consumer-grade standalone HMDs are easier to promote, but their computing power and sensing accuracy are limited. Furthermore, while HMDs can effectively induce anxiety for training, they also risk causing stronger visual fatigue and cybersickness. Researchers recommend using multiple short-duration practice sessions rather than a single long session to help participants gradually adapt to the virtual environment (Chen et al., 2025).

Regarding assessment validity and application, the systems' scientific rigor and generalizability still need to be strengthened. Standardized questionnaires and their underlying models are often established based on specific cultural backgrounds, which can lead to cultural biases or inapplicability to specific groups, resulting in skewed assessment outcomes. Furthermore, most studies only validate short-term performance gains, lacking long-term tracking of participants in actual interviews or final employment outcomes, which limits the broad applicability of the conclusions.

Regarding ethics and fairness, technology risks exacerbating social inequality. The most prominent issue is the bias in LLMs. Research confirms that LLMs used as scorers can produce discriminatory outputs based on a participant's nationality, gender, or religion (Nofal et al., 2025). Furthermore, while these systems can induce anxiety or shame, not all platforms have adequate safeguards for participants' psychological safety, potentially harming vulnerable individuals such as those on the autism spectrum (Bozgeyikli et al., 2016). Similarly, concerns have been raised that assessment systems relying heavily on non-verbal behavioral analysis may systematically bias against candidates with physical disabilities that affect their speech, movement, or facial expressions (Hickman et al., 2022). Finally, data privacy and security face severe challenges. The collection, storage, and use of sensitive information such as user identities and biometrics must be accompanied by strict ethical norms and encryption methods.

Future development must actively advance technical innovation while remaining soberly aware of these hurdles, striving to build a more reliable, equitable, and responsible next-generation of AI-driven VR interview training systems.

4.4 Future research directions

While this review leveraged database-controlled vocabularies and Google Scholar's full-text indexing to maximize paper retrieval, the search string primarily relied on broad conceptual terms like "AI." This introduces a specific limitation, as technical papers that use highly specialized nomenclature without referencing these overarching concepts might be overlooked. Future research could expand upon this work by utilizing more specific algorithmic keywords to capture further isolated sub-fields.

Evaluating the external validity of the synthesis itself is another key methodological refinement. It should be clarified that the proposed framework is a conceptual synthesis systematically derived from literature. In the absence of empirical validation through expert consultations, a Delphi consensus, stakeholder reviews, or other validation mechanisms, this framework is not intended to establish a rigid architectural standard. Instead, it serves as a structured hypothesis and an open reference design. By framing it openly, this framework can guide future system development while providing a theoretical roadmap for subsequent validation.

Beyond these methodological refinements, the technical evolution of the system itself presents critical directions. Future research should strive to evolve from passive reaction to active intervention, where the core challenge lies in identifying unconscious signals that reflect participants' true internal state. Tian et al. (2019) suggest further exploring deep correlations between multimodal data (e.g., eye-tracking features, physiological signals, and body posture) and the participants' self-efficacy and anxiety levels, thereby providing a scientific basis for the system's active intervention. The future system should not generate problems randomly. Future systems should move away from random question generation in favor of drawing on psychological exposure therapy to construct a rigorous framework for anxiety induction and tolerance training. Based on real-time monitoring of the participants' stress levels and emotional state, the system should facilitate the stimulus level gradually from weak to strong. For instance, when the system detects that a participant is under excessive stress, the virtual interviewer should switch to soothing questions, returning to challenging segments only after the participant calms down (Migovich et al., 2021). This tiered, personalized intervention based on real-time physiological feedback will enable the evolution of VR interview training from basic skill practice into psychological adaptability training. Furthermore, a critical frontier for future designs is the enhancement of ecological validity by introducing virtual interviewers with diverse personalities (e.g., cold or irritable traits) and environmental distractions including phone interruptions, background noise, or the interviewers' temporary departures. By creating interaction scenarios with situational variables and social pressure, systems can more effectively foster participants' psychological resilience and adaptive responsiveness in complex real-world settings.

To address the nationality, religious, and gender biases exhibited by LLMs in interview assessments, a critical future direction is to identify and eliminate these subconscious biases to establish fairer automated interview evaluation benchmarks. Nofal et al. (2025) propose that future research should focus on developing a bias-aware prompt system and fine-tuning LLMs to create a set of inclusive objective evaluation benchmarks that transcend traditional demographic characteristics, ensuring the system does not transmit erroneous workplace biases to participants. Furthermore, current mainstream commercial facial recognition models are trained on data from non-autistic populations, making it difficult to accurately interpret the atypical facial expressions of autistic participants, which results in significant assessment validity bias (Adiani et al., 2022). Consequently, developing test datasets specifically tailored for the neurodiverse population is the top priority of the future technological path. True fairness in the assessment process can only be achieved by integrating emotion analysis models with group adaptability into the AI evaluation system.

Currently, most studies remain confined to short-term performance evaluations within laboratory settings, primarily focusing on skill acquisition and anxiety reduction over a single session or a few weeks. These studies struggle to comprehensively measure the skill transfer effects of

participants facing complex and variable real-world workplace challenges. Therefore, larger-scale longitudinal tracking studies should be conducted in the future, with the core objective of narrowing the validity gap between the simulation environment and the real workplace. This requires researchers not only to observe performance differences in real interviews after training but also to extend the research scope to long-term employment outcomes, including hiring rates, salary levels, and job retention after employment. At the same time, long-term systematic interventions may lead to the homogenization of interview responses. Participants may tend to seek algorithmic patterns in order to achieve high scores, rather than truly enhancing their professional skills (Gupta, 2025). Thus, future system designs must strike a balance between training effectiveness and the preservation of individual expression. The long-term feedback mechanism established under this balanced framework not only verifies the actual skill transfer rate of VR training in the real market but also drives the adaptive refinement and rapid iteration of interview systems through multi-dimensional career adaptability evaluations, enabling the test participants to truly transition from virtual success to career competence.

5 Conclusion

This review synthesizes the latest research progress on AI-driven VR interview training systems, focusing on three core research questions: RQ1: how AI drives VR interview training systems to evolve from early preset scripts to adaptive interaction; RQ2: how multimodal data fusion enables comprehensive assessment of external performance and internal state; and RQ3: what challenges exist in technical efficiency, assessment validity, and ethical fairness. This review reveals a clear evolutionary path from rule-based systems and machine learning to GenAI. This process is defined by a three-layer architecture made up of Interaction, Multimodal Data Acquisition, and Algorithm modules, which provides the basis for evaluating subjective perception, objective behavior, and internal physiology. Furthermore, we identify critical challenges and research gaps, including hardware thresholds, algorithmic bias, data privacy, and longitudinal validity.

Studies indicate that the maturity of VR hardware around 2019 established the technical foundation for high-precision data collection, while the advancement of GenAI at the end of 2022 pushed the system to a new level of personalization and context-awareness. Current systems have achieved a leap from mechanical responses to intelligent perception and can provide personalized feedback based on multimodal data. However, this rapid technological evolution is inevitably accompanied by systematic challenges. Hardware costs and operational complexity continue to limit the accessibility of this technology. Furthermore, the biases regarding nationality, religion, and gender exhibited by LLMs in scoring threaten the fairness of assessments. There is also a lack of standardized privacy protocols for the acquisition and storage of multimodal data. Finally, the validity verification of most studies remains confined to laboratory environments, lacking long-term longitudinal tracking of real employment outcomes in the actual job market.

In the future, the core mission of this field lies in achieving an evolution from passive response to active intervention, which requires researchers to deeply explore the underlying correlations between multimodal data (e.g., eye-tracking features, physiological signals, and body posture) and the participants' internal states. Meanwhile, to address the algorithmic biases of LLMs, it is necessary to develop bias-aware prompt systems and dedicated test datasets tailored for neurodiverse populations, ensuring that technology serves special groups facing employment barriers with equity. Most importantly, future research must break through the limitations of short-term laboratory assessments and conduct large-scale longitudinal tracking studies to verify the actual hiring rates in the real-world market.

In summary, an AI-driven VR interview system should be more than a tool for skill practice. It should evolve into a comprehensive platform that promotes inclusive employment and cultivates psychological resilience. Only through the double drive of technological innovation and ethical reflection can we develop a new generation of intelligent systems that are more transparent, reliable, and responsible. In this way, every breakthrough achieved within a virtual interview can serve as a solid stepping stone toward tangible career opportunities.

Data availability statement

The original contributions presented in the study are included in the article/Supplementary Material, further inquiries can be directed to the corresponding authors.

Author contributions

RL: Writing – original draft, Investigation, Writing – review and editing, Conceptualization, Methodology. RY: Writing – review and editing, Supervision, Methodology, Conceptualization.

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The author(s) declared that generative AI was not used in the creation of this manuscript.

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Supplementary material

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